Real affordability in major U.S. metros

Final Presentation

WINTER 2026

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Our key question:

How has real housing affordability changed across major U.S. cities from 2015–2024, and which cities are the most and least "affordable"?

- Target Audience:
  - Young professionals choosing a place to live/work
  - Federal, state, and local leadership looking to tackle affordability issues
- Why they care:
  - Inflation means that **REAL** affordability is disconnected from nominal prices
  - In uncertain times, affordability is a critical element of the attractiveness of cities

Our base metrics (from U.S. Census data & Federal Reserve Economic Data:

Real housing affordability

- CPI-adjusted median rent
- CPI-adjusted median home prices
- Housing cost burden
- Nominal & real median income

Macroeconomic factors

- Interest rates
- Unemployment
- Mortgage delinquencies
- Education level

U.S. housing affordability is a regional issue

We compared median nominal home prices with "real" (CPI-adjusted) home prices. This standardizes home values across the U.S. by using regional CPI.

City	Median Home Value	Real Home Value	CPI Index
San Francisco	$1,199,500.00	$1,030,511.04	297
Seattle	$773,011.11	$658,135.49	287
Los Angeles	$749,611.11	$642,367.06	277
New York	$663,900.00	$601,038.44	285
Boston	$608,744.44	$538,678.22	288
Atlanta	$352,277.78	$299,608.81	256
Chicago	$283,977.78	$256,860.02	251
Dallas	$244,366.67	$210,548.74	248
Houston	$216,855.56	$191,737.42	237
Philadelphia	$197,844.44	$178,735.61	267

Net growth gap shows where real income gains are outpacing housing burden

Net growth gap is a proxy for purchasing power, and is calculated by subtracting the growth in housing burden from income growth.

City	Income Growth Pct	Burden Growth Pct	Net Growth Gap
Atlanta	24.4%	7.1%	17.3%
Chicago	24.8%	-2.0%	26.8%
Boston	28.4%	-2.7%	31.0%
New York	14.8%	-3.4%	18.2%
Los Angeles	16.0%	-3.6%	19.7%
San Francisco	12.1%	-1.2%	13.3%
Philadelphia	14.5%	-2.8%	17.3%
Dallas	17.6%	5.2%	12.3%
Houston	3.8%	13.3%	-9.5%
Seattle	3.9%	-0.2%	4.1%

"Misery Index" indicates the economic distress felt by the typical household

"Misery Index" is calculated by adding the CPI and unemployment rate. In our analysis, most of the "misery" is derived from the CPI.

City	Mean CPI Index	Mean Unemployment Rate	Misery Index
San Francisco	296.51	2.98	299.49
Boston	288.38	3.62	292.00
Seattle	286.73	2.77	289.51
New York	284.82	3.62	288.44
Los Angeles	277.40	3.79	281.19
Philadelphia	267.29	4.18	271.47
Atlanta	256.06	3.30	259.36
Chicago	250.74	4.33	255.07
Dallas	247.76	2.55	250.31
Houston	236.54	3.29	239.83

Ratio of real house value to median income reveals affordability issues

The real house value-to-income ratio shows how many years of median salary it would take to afford a median home in a given market.

City	Mean (2015-2024)	Min (2015-2024)	Max (2015-2024)
Los Angeles	11.09118	10.34145	11.86971
San Francisco	10.12388	9.404082	10.72349
New York	9.81525	9.585365	10.11144
Boston	7.855266	7.258772	8.320825
Seattle	7.593119	6.607425	8.110739
Atlanta	5.052073	4.624684	5.393282
Chicago	4.506883	4.232568	4.745099
Dallas	4.155863	3.31896	4.899083
Philadelphia	3.967447	3.636482	4.209353
Houston	3.900615	3.166611	4.679853

Ratio of income to rent shows differences between renting and buying

Ratio of rent to income shows how many months of median income covers median rent across our metro areas.

City	Months of Rent Covered by Income
Seattle	59
San Francisco	58
Chicago	53
Atlanta	51
Dallas	49
Houston	49
Boston	44
New York	44
Philadelphia	43
Los Angeles	42

Differences in growth rates between rent and income shows which cities are become more affordable

By subtracting average annual income growth from average annual rent growth, we can identify in which metros rent is outpacing incomes. Lower numbers are preferred.

City	Avg Annual Rent Growth Pct	Avg Annual Income Growth Pct	Annual Rent/Income Gap Pct
Chicago	5.3	6.2	-0.8
Boston	5.7	6.4	-0.7
New York	4.3	4.8	-0.5
Seattle	4.8	5.2	-0.4
Los Angeles	5.9	6.1	-0.2
Atlanta	7.3	7.4	-0.1
San Francisco	4.7	4.6	0.1
Philadelphia	6.4	5.7	0.6
Dallas	8	6.7	1.3
Houston	5.8	4.4	1.4

By multiplying percent of population with a bachelors and rental affordability, we found where education meets affordability

This shows us places that are affordable but also present a strong market for college graduates.

City	Avg Bachelors Pct	Months of Rent Covered by Income	Education Affordability Score
Seattle	66%	59	39
San Francisco	59%	58	34
Atlanta	55%	51	28
Boston	52%	44	23
Chicago	42%	53	22
New York	40%	44	17
Dallas	35%	49	17
Houston	34%	49	17
Los Angeles	36%	42	15
Philadelphia	32%	43	14

Our synthesis: aggregate score

Polarization: High cost/high income vs. low cost/low income

We developed an aggregate score by assigning weights to each of our 7 rankings:

- Real home value = 0.15
- Purchasing power = 0.15
- "Misery Index" = 0.1
- Income to home price ratio = 0.15
- Rent to income ratio = 0.15
- Rent growth vs. income growth = 0.15
- Education affordability = 0.15

City	Weighted Final Score
Chicago	3.15
Atlanta	4.3
Dallas	4.925
Boston	5.025
Seattle	5.3
Houston	5.575
Philadelphia	6.05
New York	6.325
San Francisco	6.7
Los Angeles	7.65

Note: Lower score is better

Our synthesis: aggregate score

Takeaways:

Conclusion from analysis on affordability indices:

- Price vs. Value: Low home prices do not guarantee affordability. HOU / PHI have "cheapest" housing, but lack rent coverage
- The "best" markets succeed by balancing costs with solid income growth and perform well in multiple categories

City	Weighted Score	Home Value Rank	Purchase Power Rank	Misery Rank	Growth Gap Rank	Income Growth Rank	Rent Coverage Rank	Education Score Rank
Chicago	3.15	4	4	3	1	2	3	5
Atlanta	4.3	5	5	4	6	3	4	3
Dallas	4.925	3	3	2	9	4	5.5	7
Boston	5.025	6	7	9	2	1	7.5	4
Seattle	5.3	9	6	8	4	9	1	1
Houston	5.575	2	1	1	10	10	5.5	8
Philadelphia	6.05	1	2	5	8	7	9	10
New York	6.325	7	8	7	3	6	7.5	6
San Francisco	6.7	10	9	10	7	8	2	2
Los Angeles	7.65	8	10	6	5	5	10	9

Note: Lower score is better

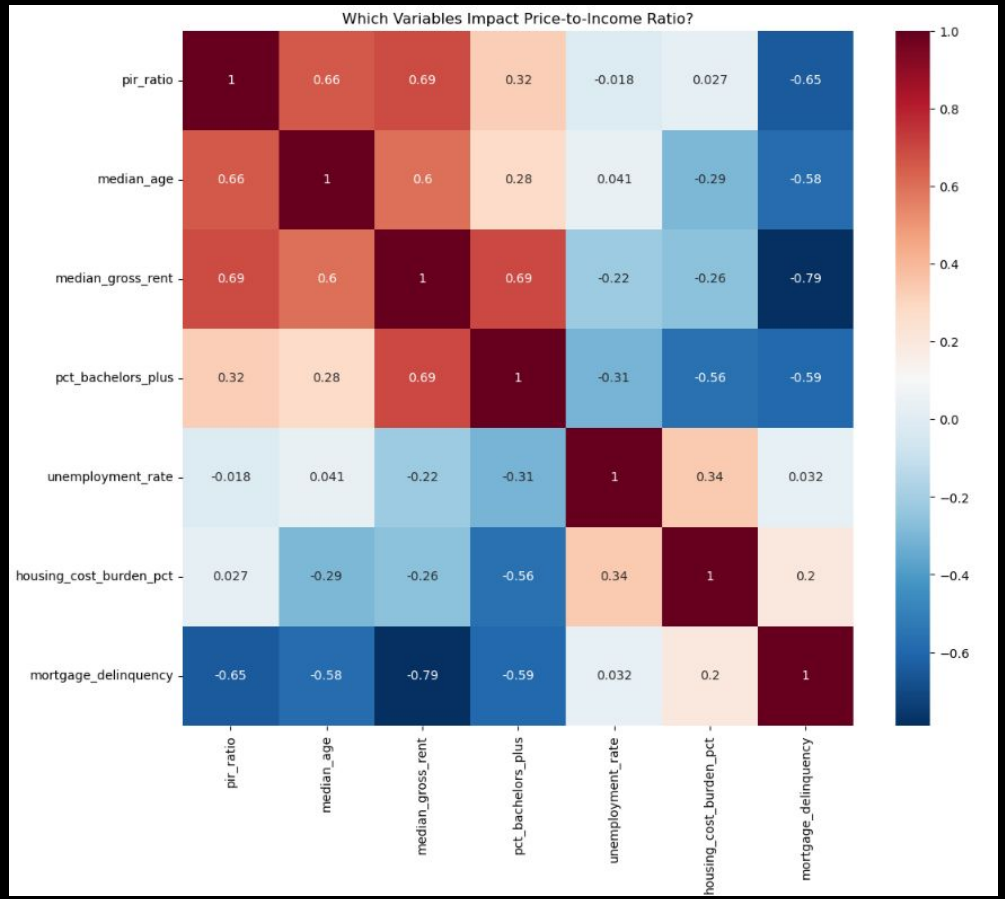
Our linear regression: predicting home price-to-income ratio

Coefficients

- Median Age = 0.38: every one-year increase in the median age of a city, adds 0.29 to the price-to-income ratio. This suggests that "older" cities tend to be less affordable.
- Median Gross Rent = 0.004: $250 rent increase adds 1 point to PIR.
- % Bachelors = -0.055: For every 1% increase in residents with a bachelor's degree, the PIR decreases by 0.05. This suggests that higher education levels correlate with better affordability, likely because incomes rise faster than home prices in these areas.
- Mortgage Delinquency = -0.77: For every 1% increase in mortgage delinquency, the PIR decreases by 0.77. This indicates that economic distress or higher risk in a market generally suppresses home price premiums.

R-squared value is .604. This means that our independent variables explain ~60% of the variance in Real Home prices.

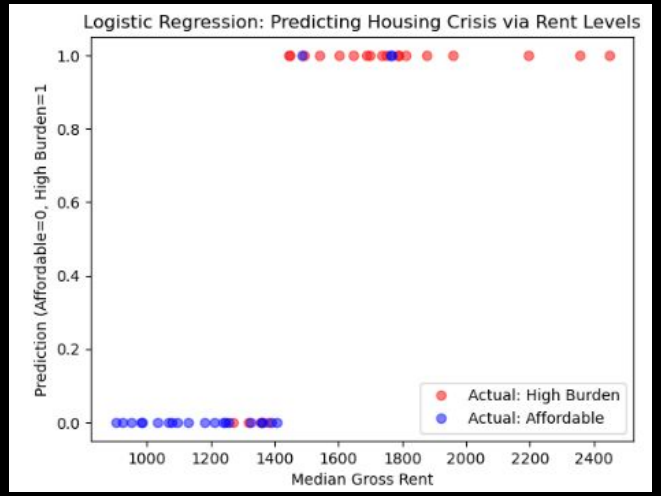
RMSE: On average, our model's RMSE is 1.75 (years of income required to purchase a house in each city). The average city in our data has a ratio of 6.81.

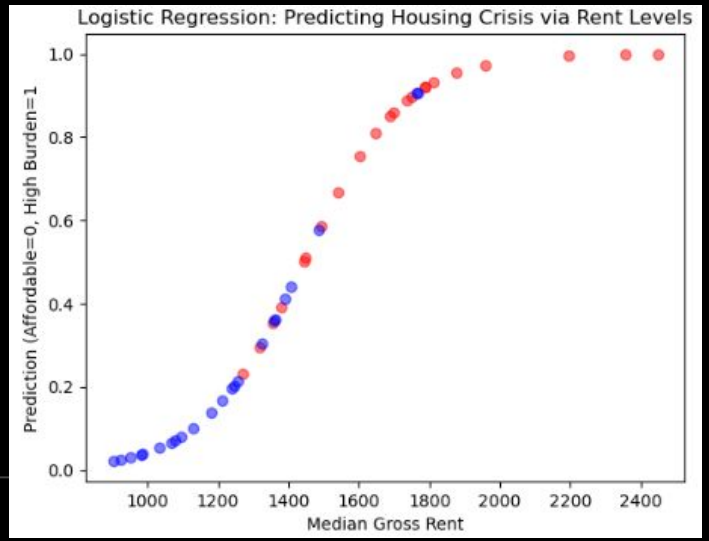


Our logistic regression: predicting a housing crisis

We calculated the whether a city is in a high-burden state (Price-to-Income ratio > 6.0).

- Our model achieved an **accuracy of 80%** using median gross rent (initially included median age, pct bachelors plus, and mortgage delinquency; however, removed variables following analysis of pair plots).
- Our confusion matrix shows that the model accurately identified **20 affordable and 18 high-burden cities.**
    - It was reliable; however, did produce **3 false positives and 4 false negatives.**
- **Next Steps:** A larger pool of data in the form of many more than 10 cities (and more than 10 years worth of data) would provide better training and testing data for our model.





Conclusion / Questions